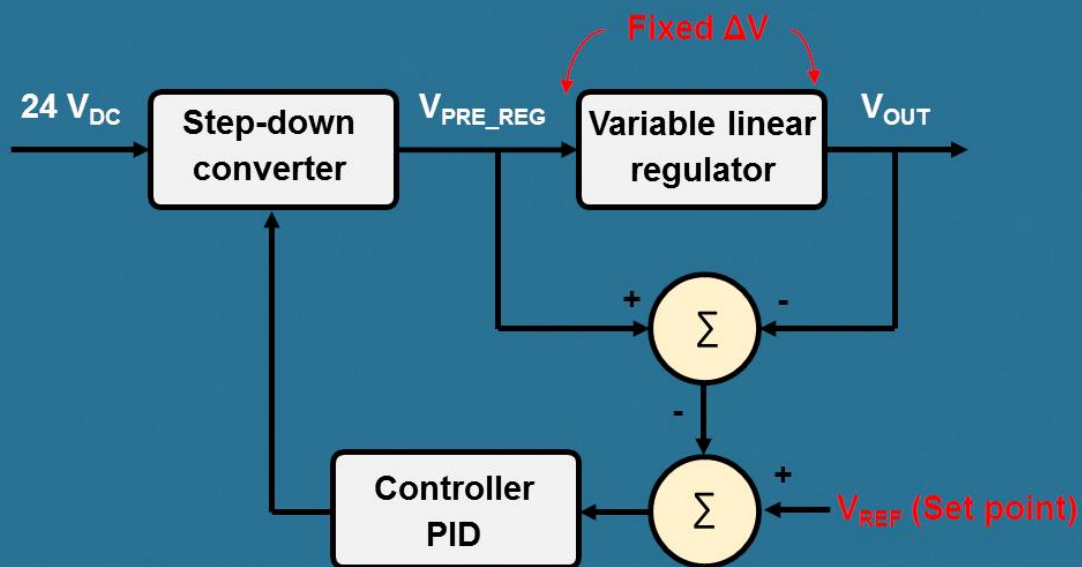


Design of a hybrid variable power supply



Leonardo Chieco

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The design of variable power supplies is one of the most interesting topics in power electronics, as it requires balancing energy efficiency, output voltage stability, low thermal dissipation, and circuit simplicity. In this tutorial, a hybrid solution combining a switching converter based on the LM2596-ADJ controller with an LM317 linear regulator will be analyzed, taking advantage of the strengths of both technologies to achieve a highly efficient variable power supply.

The main objective is to understand how the power dissipated by the linear regulator can be drastically reduced by using a switching pre-regulator capable of dynamically adapting its output voltage to the actual requirements of the load. This approach makes it possible to maintain high overall system efficiency without sacrificing the regulation quality typically associated with linear power supplies.

The tutorial has been developed with a primarily educational approach. As a baseline framework, the project leaves ample room for improvement. For instance, it currently lacks the standard protection mechanisms typically required in an industrial environment.

To support the theoretical analysis, the design will be verified through circuit simulation using LTspice. The obtained results will allow observation of the system's dynamic behavior and verification of the correct operation of the regulation loop.

Enjoy the reading!

About the Author

Leonardo Chieco is an electronic engineer with more than 20 years of experience in the design and development of automation control software (PC/PLC), electronic boards for industrial applications, firmware, robotics, and mechatronics.

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1. Introduction

The design of variable voltage power supplies often requires balancing seemingly conflicting requirements. On one hand, it is desirable to achieve high energy efficiency by minimizing power losses and thermal dissipation. On the other hand, the output voltage is expected to exhibit low noise, high stability, and strong rejection of disturbances originating from the input supply.

Pure switching and pure linear architecture offer complementary advantages and disadvantages.

A switching regulator can achieve efficiencies exceeding 90%, since the power device, typically a MOSFET, operates mainly in either cutoff or saturation, thereby dissipating a relatively small amount of energy. However, high-frequency switching inevitably introduces ripple, electromagnetic interference (EMI), and spectral components that may be undesirable in laboratory or sensitive analog applications.

A linear regulator, by contrast, can provide an extremely clean and stable output voltage, but this benefit comes at the expense of significant power dissipation. If a linear regulator must reduce an input voltage of 24 V down to only a few volts while delivering a relatively high current, the amount of power converted into heat can become substantial. The power dissipated by the regulating device is given by the well-known relationship:

$$P_D = (V_{in} - V_{out}) \cdot I_{out}$$

From this equation, it immediately follows that in the case of a linear power supply operating from 24 V and regulated to 10 V at an output current of 1 A, the regulator dissipates 14 W. Such a value requires a relatively large heatsink and results in a reduction of the overall system efficiency. These considerations become even more important in battery-powered equipment or whenever compact dimensions are required.

To overcome this limitation, a hybrid architecture can be adopted in which a switching converter performs a first coarse voltage conversion, while a linear stage provides fine regulation and suppresses the residual ripple. This approach makes it possible to combine the strengths of both technologies simultaneously: the efficiency of switching conversion and the output quality of linear regulation.

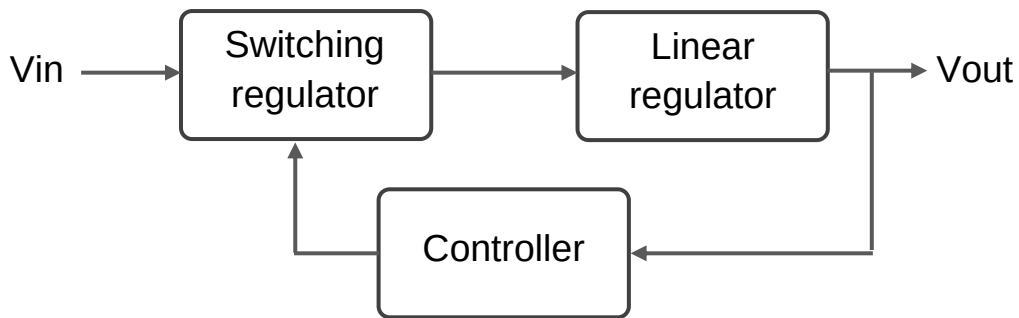


Figure 1: System Architecture

In the design described in this tutorial, a buck converter based on the LM2596-ADJ integrated circuit acts as the pre-regulator, while an LM317 provides the final voltage regulation. An additional control system dynamically adjusts the operating point of the switching converter to always maintain only a small voltage difference between the input and output of the linear regulator, thereby minimizing its power dissipation.

2. LM2596 Buck Converter Design

The LM2596 integrated circuit is one of the most widely used monolithic buck converters in both hobbyist and professional applications thanks to its ease of use, the small number of external components required, and its ability to deliver output currents of up to 3 A.

The LM2596 belongs to the family of step-down converters, that is, converters capable of reducing an applied DC input voltage. Unlike a linear regulator, which dissipates the voltage difference between input and output as heat, a buck converter transfers energy to the load through a high-frequency switching process, achieving efficiencies that can exceed 80–90% depending on operating conditions.

The operating principle is based on the periodic switching of an internal power transistor. When the transistor is conducting, energy is transferred from the supply to the inductor and the load. When the transistor turns off, the energy stored in the inductor's magnetic field continues to supply the load through the freewheeling diode. The filter formed by the inductor and output capacitor converts the sequence of switching pulses into a relatively stable DC output voltage.

Under ideal conditions, the output voltage of a buck converter can be described by:

$$V_{out} = D V_{in}$$

where D is the duty cycle, i.e., the fraction of the switching period during which the internal transistor remains on.

The LM2596 operates at a typical switching frequency of approximately 150 kHz, representing a good compromise between inductor size, efficiency, and circuit complexity. An internal control loop continuously monitors the voltage present at the feedback pin. When the output voltage tends to decrease due to an increased load, the controller automatically increases the duty cycle to transfer more energy to the load. Conversely, if the output voltage rises above the desired value, the duty cycle is reduced.

The output voltage is established through a feedback network connected to the FB pin. In the adjustable version of the LM2596, the controller maintains an internal reference voltage of approximately 1.23 V at the FB pin. Consequently, the output voltage is determined by the resistor divider ratio according to:

$$V_{out} = 1.23 \cdot \left(1 + \frac{R_2}{R_1}\right)$$

where R_1 is connected between the FB pin and ground, while R_2 is connected between the converter output and the FB pin.

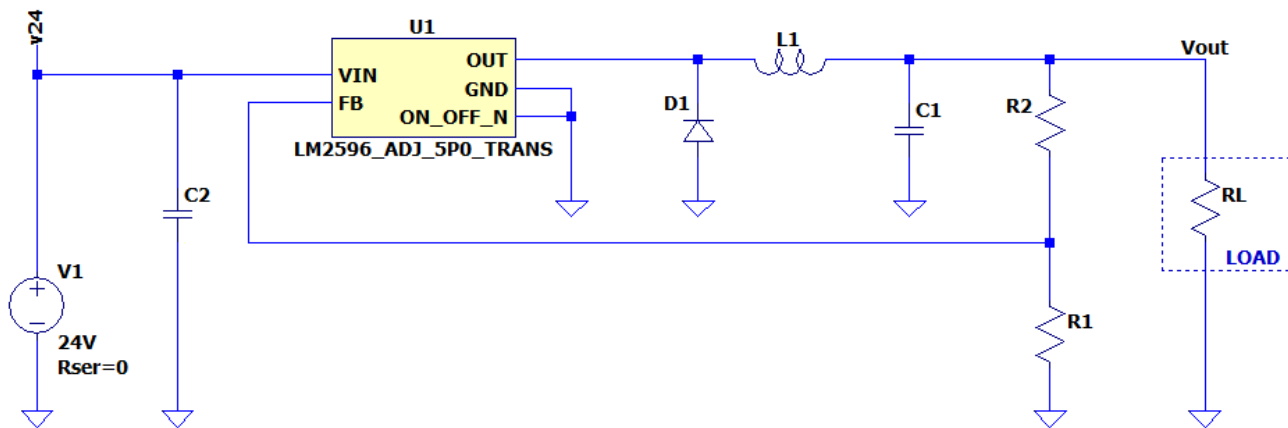


Figure 2: Typical Application Circuit of the LM2596-ADJ

In the circuit presented in this tutorial, the resistor divider, initially designed to provide a minimum output voltage of approximately 3 V, will be dynamically reconfigured to vary its division ratio, allowing the output voltage to be adjusted up to 22-23 V.

For the moment, let us assume that the converter operates at half the supply voltage, namely 12 V.

For the converter design, we will follow the procedure recommended in the device datasheet and adapt it to the specific requirements of our application.

Design Specifications:

- Input voltage: $V_{in} = 24 \text{ V}$
- Output voltage: $V_{out} = 12 \text{ V}$

- Maximum output current: $I_{out} = 2\text{ A}$
- Switching frequency: $f_s = 150\text{ kHz}$

We can now calculate the remaining component values.

Feedback resistors

For the adjustable version, the datasheet provides:

$$V_{out} = V_{ref} \cdot \left(1 + \frac{R_2}{R_1}\right)$$

With:

$$V_{ref} = 1.23\text{ V}$$

A common choice is: $R_1 = 1\text{ k}\Omega$, therefore:

$$R_2 = R_1 \cdot \left(\frac{V_{out}}{V_{ref}} - 1\right)$$

$$R_2 = 1000\Omega \cdot \left(\frac{12\text{ V}}{1.23\text{ V}} - 1\right) = 8.7\text{ k}\Omega$$

Calculation of E·T product

The datasheet defines:

$$E \cdot T = \frac{(V_{in} - V_{out} - V_{sat})(V_{out} + V_D)}{V_{in} - V_{sat} + V_D} \cdot \frac{1000}{150\text{ kHz}}$$

where:

- $V_{sat} = 1.16\text{ V}$
- $V_D = 0.5\text{ V}$

These are typical values reported in the datasheet.

Substituting:

$$E \cdot T = \frac{(24 - 12 - 1.16)(12 + 0.5)}{24 - 1.16 + 0.5} \cdot \frac{1000}{150\text{ kHz}} = 38.7\text{ V}\mu\text{s}$$

Inductor Selection

According to the datasheet selection curves, an E·T value of approximately 39 V·μs at an output current of 2 A falls within the region corresponding to a 68 μH inductor.

The inductor should also satisfy the following requirements:

- Rated current > 2 A (recommended ≥ 3 A)
- Saturation current ≥ 4 A
- Lowest possible DCR (Direct Current Resistance)

Let us examine these parameters in more detail.

1- DCR is simply the ohmic resistance of the copper winding.

For example, if the inductor has a DCR of 50 mΩ and carries 2 A:

$$P = R_{DCR} I^2 = 0.05\Omega (2A)^2 = 0.2 W$$

which is entirely dissipated as heat.

If the DCR increases to 200 mΩ:

$$P = R_{DCR} I^2 = 0.2\Omega (2A)^2 = 0.8 W,$$

four times higher.

For this reason, switching power supplies always aim for the lowest possible DCR to achieve lower heating, higher efficiency, reduced voltage drop, improved regulation;

For a 2 A LM2596 buck converter, a DCR below 50 mΩ is an excellent choice.

2- The rated current specifies the maximum continuous current that can flow through the inductor without exceeding a temperature rise defined by the manufacturer.

For example:

- Rated Current = 3 A
- Temperature Rise = 40°C

This means that with 3 A continuously flowing, the inductor reaches a temperature considered acceptable by the manufacturer.

Exceeding this value generally does not cause immediate failure, but:

- the component operates at a higher temperature,
- thermal stress increases,
- long-term reliability may decrease.

3- The inductor operates thanks to the magnetic field generated in its core. When the current becomes too high, the core enters saturation and the inductance rapidly collapses. As a result:

- Current ripple increases,
- Converter stability deteriorates,
- The LM2596 internal switch may be subjected to excessive current stress.

The saturation current is therefore often the most critical parameter.

For our design $V_{in} = 24\text{ V}$, $V_{out} = 12\text{ V}$, $I_{out} = 2\text{ A}$

The average inductor current is approximately equal to the output current:

$$I_L \approx I_{out}$$

However, the current waveform is triangular.

Assuming a ripple current of 30%:

$$\Delta I_L = 0.3 \cdot 2\text{ A} = 0.6\text{ A}$$

The peak current becomes:

$$I_{PK} = I_{out} + \frac{\Delta I_L}{2} = 2.3\text{ A}$$

Therefore, the core must support at least 2.3 A without saturation.

A safety margin of 30–50% is normally applied.

Using a 50% margin:

$$I_{SAT} = I_{PK} \cdot 1.5 = 2.3 \text{ A} \cdot 1.5 = 3.45 \text{ A}$$

which leads to the recommendation: $I_{SAT} \geq 4\text{A}$

Output Capacitor

For the LM2596, the datasheet recommends low-ESR electrolytic capacitors.

For 12 V operation and currents up to 2 A, a typical value is:

$$C_{out} = 330 \mu\text{F}$$

With 25 V rated voltage and low ESR.

To further reduce ripple, a 100 nF ceramic capacitor is usually connected in parallel.

Input Capacitor

For a 24 V input and output currents up to 2 A, the datasheet typically recommends:

$$C_{in} = 220 \mu\text{F}$$

with a voltage rating of at least 35 V.

As before, a 100 nF ceramic capacitor should be placed close to the regulator pins.

Schottky diode

The average diode current is approximately:

$$I_D \approx I_{out} (1 - D)$$

with duty cycle $D = \frac{12}{24} = 0.5$.

Therefore:

$$I_D \approx 2 (1 - 0.5) = 1A.$$

The datasheet recommends a generous safety margin and suggests Schottky diodes rated for at least 3 A.

Feed-Forward Capacitor

The LC output filter of a buck regulator introduces a phase drop (up to 180°) near its resonant frequency. If the total loop phase shift reaches 360° where the gain is 1 (0 dB), the regulator becomes unstable and oscillates.

For this reason, the LM2596-ADJ datasheet suggests adding a feed-forward capacitor in parallel with R_2 .

By placing C_f across R_2 , higher-frequency AC signals bypass the resistor network and are fed directly into the feedback (FB) pin. This injects phase lead (extra phase margin) into the feedback loop before the gain crosses 0 dB, preventing oscillation and stabilizing the system, especially when the output capacitor has ultra-low ESR and cannot provide enough stability on its own.

For the LM2596-ADJ, a good rule of thumb for calculating is based on the target zero frequency f_z , typically chosen between close to 5kHz.

The capacitor value is given by relation:

$$C_f = \frac{1}{2 \pi R_2 f_z} = \frac{1}{6.28 \cdot 8700 \cdot 5000} \approx 3.9 \text{ nF}$$

The following image shows the simulation of the complete circuit.

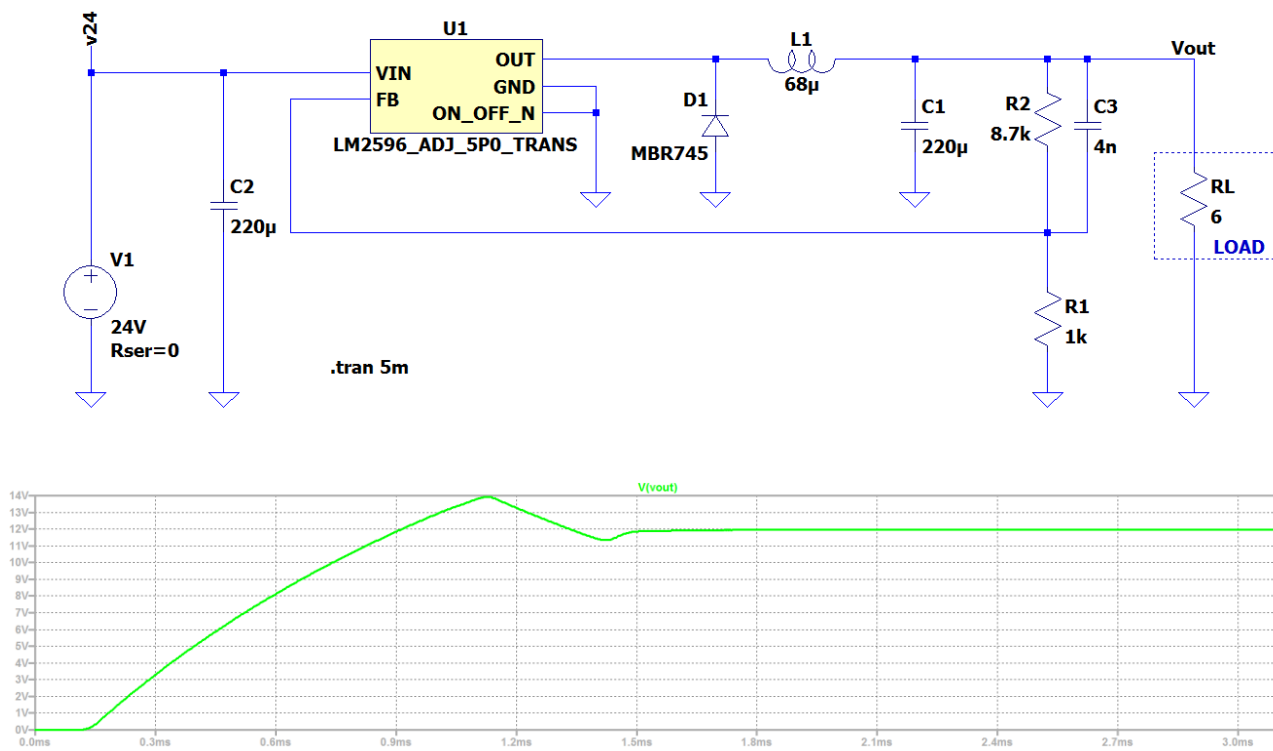


Figure 3: Operation of the LM2596-ADJ-Based Circuit

For additional details regarding component selection, layout recommendations, and stability considerations, please refer to the device datasheet.

3. Linear regulator LM317

The LM317 integrated circuit is probably one of the best-known and most widely used adjustable linear voltage regulators in analog electronics. Its popularity stems from its ease of use, robustness, and ability to provide an adjustable output voltage using only two resistors.

Its operating principle is based on maintaining a constant voltage between the OUT and ADJ terminals. This voltage is approximately 1.25 V. By using a resistive voltage divider, it is possible to set the desired output voltage value.

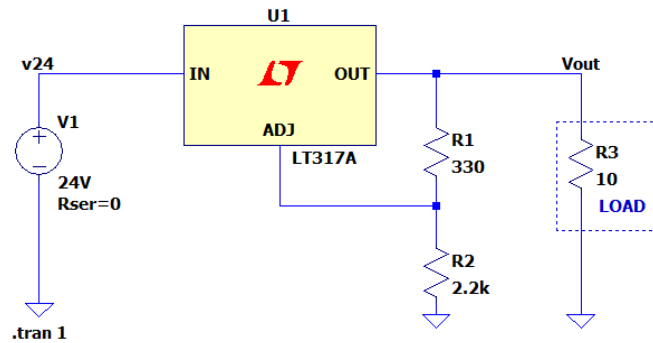


Figure 4: Typical application circuit of the LM317 integrated regulator

The fundamental relationship is:

$$V_{out} = 1.25 \cdot \left(1 + \frac{R_2}{R_1} \right) + I_{ADJ} \cdot R_2$$

Since the current flowing into the ADJ terminal (I_{ADJ}) is generally very small, the second term is often neglected in practical applications.

For the circuit shown in the previous figure:

$$V_{out} = 1.25 \cdot \left(1 + \frac{R_2}{R_1} \right) = 1.25 \cdot \left(1 + \frac{2200\Omega}{330\Omega} \right) = 9.6 V$$

The LM317 also provides good ripple rejection from the power supply. This characteristic makes it particularly suitable as a post-regulation stage following a switching converter. A significant portion of the high-frequency noise generated

by the switching process is attenuated by the linear regulator, resulting in a much cleaner output voltage.

The main limitation of the LM317 is its dropout voltage, the minimum differential voltage required between input and output to maintain regulation, which typically ranges from 2 V to 3 V. Consequently, the input voltage must always exceed the output voltage by at least this margin.

Furthermore, the voltage differential between the input and output multiplied by the load current represents the power that the IC must dissipate as heat. This makes thermal management a critical factor, especially when the input voltage is significantly higher than the output voltage.

4. The proposed architecture

We previously saw that the output voltage of the LM2596 regulator can be set by the feedback network composed of two resistors according to the relationship:

$$V_{out} = V_{ref} \cdot \left(1 + \frac{R_2}{R_1}\right)$$

where R_1 is connected between the FB pin and ground; R_2 is connected between the converter output and the FB pin; and $V_{ref} = 1.23V$

In this project, however, the switching converter does not operate as a simple fixed-voltage source but rather as a dynamic pre-regulator. Its task is to provide the LM317 with a voltage V_{SW} that is 5 V higher than the desired output voltage V_{out} , with the aim of minimizing the power dissipated by the linear regulator.

Assuming a desired output voltage of:

$$V_{out} = 10 V$$

the power dissipated by the regulator becomes:

$$P_D = (V_{SW} - V_{out}) \cdot I_{out} = (15 - 10) \cdot 1 = 5W$$

instead of

$$P_D = (V_{SW} - V_{out}) \cdot I_{out} = (24 - 10) \cdot 1 = 14W$$

resulting in approximately a threefold reduction in power dissipation.

The entire system can therefore be viewed as a linear power supply assisted by an intelligent switching stage. The LM2596 converter delivers energy with high efficiency, while the LM317 acts as an analog post-regulator, providing stability, low noise, and accurate voltage regulation.

Let us now examine the system design in greater detail.

Dynamic Feedback Network

The element that allows the PID controller to dynamically modify the voltage generated by the LM2596-ADJ converter consists of an interface stage formed by a PMOS transistor and an NPN transistor. The purpose of this circuit is to convert the analog control signal, generated by a PID controller that will be discussed later, into a variation of the switching converter's feedback network without altering its internal operating principle.

To understand the regulation mechanism, it is useful to recall that the LM2596-ADJ controls its output voltage by maintaining the FB (feedback) pin at a constant voltage of approximately 1.23V. In a conventional configuration, the output voltage is determined by a resistive voltage divider connected between the output and ground.

When the voltage at the FB pin tends to decrease, the regulator increases the switching duty cycle to raise the output voltage. Conversely, when the voltage at the FB pin increases, the duty cycle is reduced.

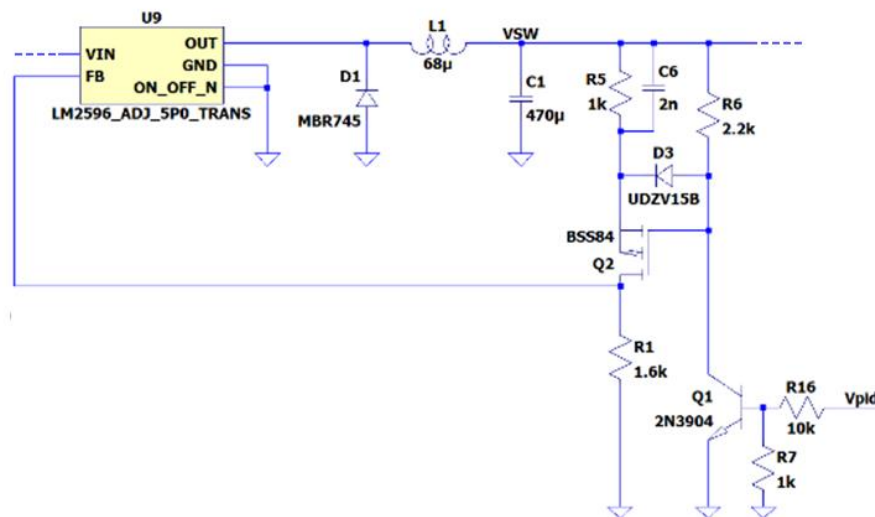


Figure 5: Dynamic feedback circuit

In the proposed circuit, the upper resistor of the feedback divider (R_5) is not connected directly to the FB pin. Instead, it is placed in series with the drain-source channel of a PMOS transistor. The PMOS operates in the linear region and therefore behaves as a voltage-controlled resistor. By varying the voltage

applied to the PMOS gate, its equivalent resistance can be modified, thereby altering the feedback divider ratio seen by the switching regulator.

The PMOS gate is normally held at the same voltage as the converter output, V_{SW} , by means of a 2.2 k Ω pull-up resistor. Under this condition, the gate-source voltage is $V_{GS} \geq 0$, and the PMOS is either cut off or exhibits a very high drain-source resistance. Referring to the previous figure, the output voltage of the LM2596-ADJ is given by:

$$V_{SW} = 1.23 \cdot \left(1 + \frac{R_5 + R_{DS,PMOS}}{R_1}\right)$$

The PMOS gate is also connected to the collector of an NPN transistor whose emitter is tied to ground. The NPN base receives the signal generated by the PID controller through a suitable resistive divider that adapts the voltage level to the transistor requirements.

When the PID output is low ($V_{PID} \leq 0$), the NPN transistor remains cut off. In this condition, the PMOS gate is held at the voltage V_{SW} by the pull-up resistor, and the PMOS exhibits a high resistance $R_{DS,PMOS}$. Consequently, the switching converter operates at its maximum programmed output voltage.

When the PID determines that the pre-regulation voltage must be reduced, the signal at the controller output increases until the NPN transistor begins to conduct. The NPN then starts sinking current from the PMOS gate node, progressively lowering its potential. As a result, V_{GS} becomes negative (the source voltage is higher than the gate voltage, which is driven close to ground), causing the PMOS to conduct. The PMOS channel gradually opens and its drain-source resistance $R_{DS,PMOS}$ decreases.

From the perspective of the feedback network, a decrease in the PMOS resistance is equivalent to modifying the upper resistor of the divider. Consequently, the fraction of the output voltage applied to the FB pin changes. The LM2596 interprets this variation as a deviation from the regulation setpoint and reacts by adjusting the switching duty cycle.

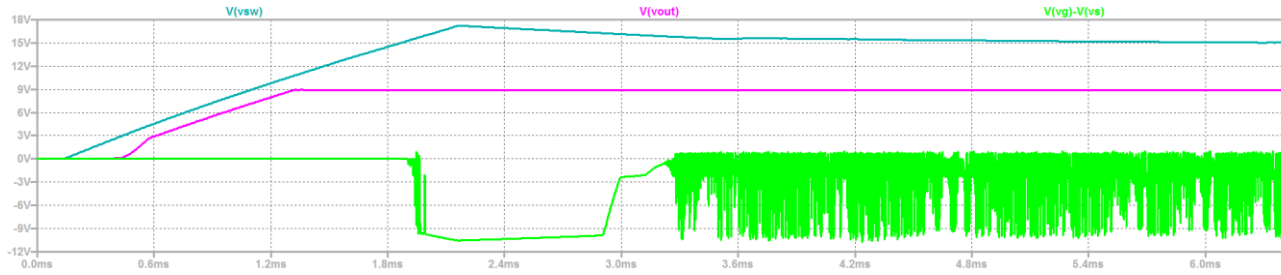


Figure 6: MOSFET V_{GS} waveform during voltage regulation

A closed-loop control mechanism is therefore established: the PID controller directly measures the difference between the desired voltage and the actual voltage, generates a corrective signal, and the NPN transistor converts this signal into a gate-source voltage for the PMOS, which modifies the feedback network of the switching converter. The converter then automatically adjusts its output voltage until a new equilibrium point is reached.

The use of the PMOS as a controlled resistor offers several advantages:

- It allows intervention on the converter feedback network without disrupting the operation of the LM2596 internal control loop.
- It enables continuous and fully analog output-voltage regulation, avoiding the nonlinearities typically associated with solutions based on discrete switching techniques.

Design of the Analog PID

The core of the entire pre-regulation system is the analog PID controller. Its purpose is to regulate the voltage produced by the LM2596 converter so that it continuously follows the relationship:

$$V_{SW} = V_{out} + 5V$$

where V_{SW} is the voltage generated by the buck converter and V_{out} is the final output voltage of the power supply. In this way, the LM317 linear regulator always operates with a voltage differential of approximately 5 V, drastically reducing power dissipation.

From a functional standpoint, the PID processes the error between a reference voltage (5 V) and a voltage equal to the difference between the switching converter output voltage and the final output voltage.

To ensure good system stability, the controller has been implemented using three independent stages dedicated to proportional, integral, and derivative actions. This approach allows the dynamic parameters of the system to be adjusted separately and greatly simplifies the analysis of the control-loop behavior.

Error Computation Stage

The voltages V_{SW} and V_{out} , after being reduced by a voltage divider with a gain of $\frac{1}{2}$, are applied to a differential amplifier stage, denoted as U4.

The output voltage of U4 is given by:

$$V_{U4} = \left(\frac{V_{SW}}{2} - \frac{V_{out}}{2} \right) \frac{R_8}{R_{11}} \approx \left(\frac{V_{SW} - V_{out}}{2} \right) \cdot 2 = V_{SW} - V_{out}$$

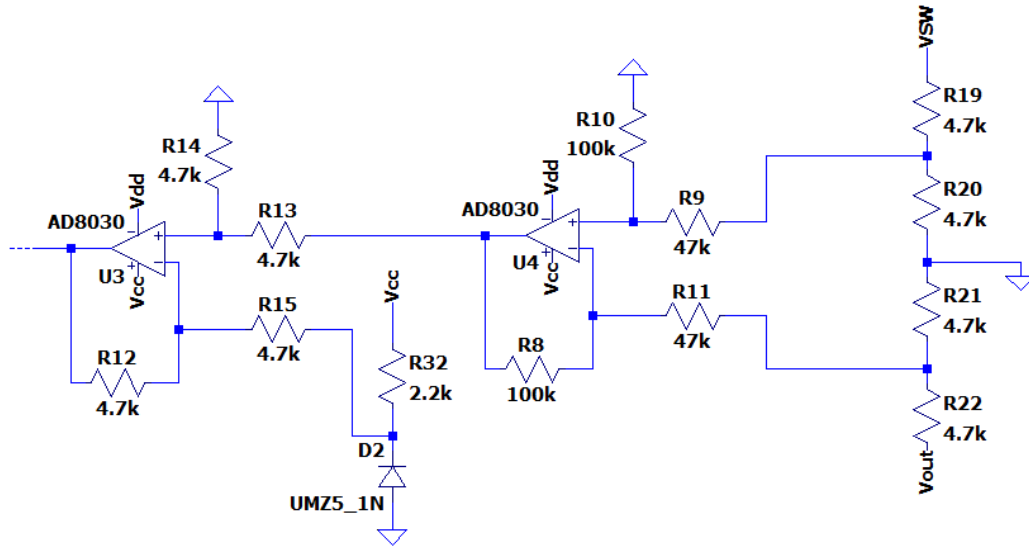


Figure 7: Error measurement stage

The stage built around U3 performs the subtraction between V_{U4} and the voltage across the Zener diode D2, which is approximately 5.4 V. Since $R_{12} = R_{15}$ and $R_{13} = R_{14}$, the output voltage is:

$$V_{U3} = (V_{U4} - V_{D2}) \frac{R_{12}}{R_{15}} \approx (V_{U4} - V_{D2}) = (V_{SW} - V_{out}) - V_{D2}$$

Therefore, V_{U3} is precisely the error signal that drives the PID controller.

When the difference $(V_{SW} - V_{out})$ is equal to the Zener reference voltage V_{D2} (approximately 5.4 V), the error signal becomes zero and the system is at its equilibrium operating point. Any deviation from this condition generates a positive or negative error signal, which is subsequently processed by the proportional, integral, and derivative stages of the controller to restore the desired voltage difference.

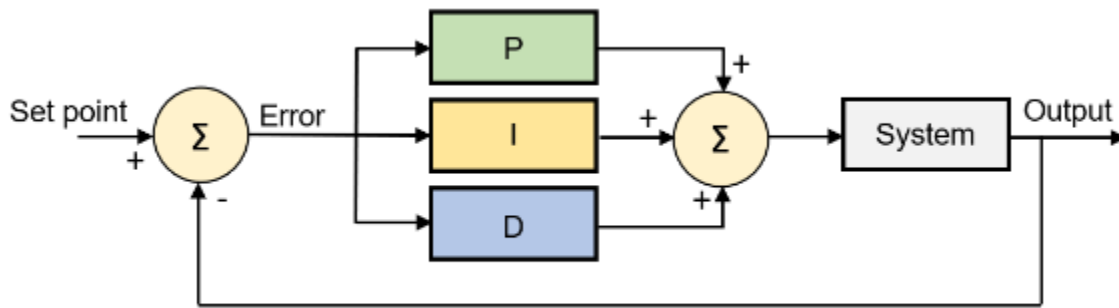


Figure 8: PID Control system

Proportional stage

The proportional stage is implemented using a non-inverting amplifier.

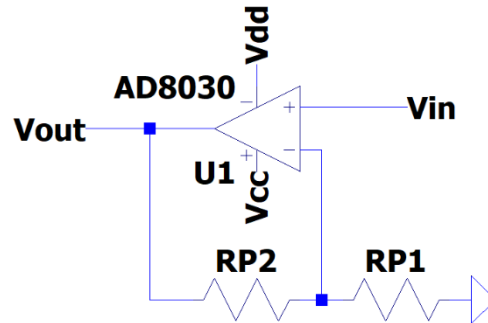


Figure 9: Proportional stage

Let R_{P1} be the resistor connected from the inverting input to ground and R_{P2} the feedback resistor. The gain is:

$$A_p = 1 + \frac{R_{P2}}{R_{P1}}$$

The purpose of this block is to generate a correction that is proportional to the error.

Assuming $R_{P1} = 4.7k\Omega$ e $R_{P2} = 12k\Omega$, the gain becomes:

$$A_p = 1 + \frac{R_{P2}}{R_{P1}} = 1 + \frac{12k\Omega}{4.7k\Omega} = 3.5$$

This means that an error variation of 100 mV produces a proportional output variation of approximately 350 mV.

A high proportional gain improves response speed but increases the likelihood of oscillations.

Integral Stage

The integral action is responsible for eliminating steady-state error.

The stage is implemented as a non-inverting integrator obtained by combining an RC network with an operational amplifier.

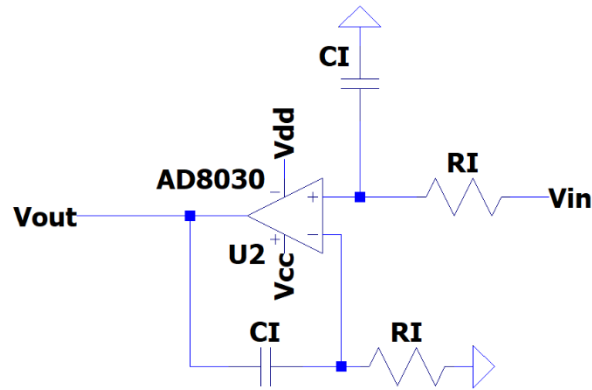


Figure 10: Integral stage

Recall that the voltages at the inverting and non-inverting inputs of an op-amp are equal and that the input currents are negligible.

Under these assumptions:

$$\frac{V_{in} - V_+}{R_I} = C_I \frac{dV_+}{dt}$$

Furthermore:

$$\frac{V_-}{R_I} = C_I \frac{d(V_{out} - V_-)}{dt}$$

Since

$$V_+ = V_- = V$$

$$\frac{V}{R_I} = C_I \frac{dV_{out}}{dt} - C_I \frac{dV}{dt}$$

Since:

$$\frac{V_{in} - V}{R_I} = C_I \frac{dV}{dt}$$

We obtain:

$$\frac{V}{R_I} = C_I \frac{dV_{out}}{dt} - \frac{V_{in} - V}{R_I}$$

Therefore:

$$C_I \frac{dV_{out}}{dt} = \frac{V_{in}}{R_I}$$

Integrating:

$$V_{out}(t) = \frac{1}{R_I C_I} \int_0^t V_{in}(t) dt$$

Assuming $R_I = 10k$ and $C_I = 47n$, we obtain:

$$K_I = \frac{1}{R_I C_I} = \frac{1}{\tau_I} = \frac{1}{10 \cdot 10^3 \cdot 47 \cdot 10^{-9}} = \frac{1}{0.00047} = 2128 \text{ s}$$

The associated time constant is:

$$\tau_I = R_I C_I = 0.00047 \text{ s}^{-1}$$

This means that the integrator progressively accumulates the error and modifies the converter control signal until the residual error is eliminated.

Larger time constants produce a more stable but slower system; smaller values increase correction speed but may introduce overshoot and oscillations.

To prevent the integrator from remaining saturated due to prolonged error accumulation (a phenomenon known as integrator windup), a resistor $R_f \gg R_I$ is generally placed in parallel with the integrating capacitor C_I . This resistor provides a low-frequency discharge path, limiting the unlimited growth of the integrated voltage and allowing the controller to recover more quickly from saturation conditions. Since R_f is much larger than R_I , the integrating behavior within the operating bandwidth of the system remains essentially unchanged.

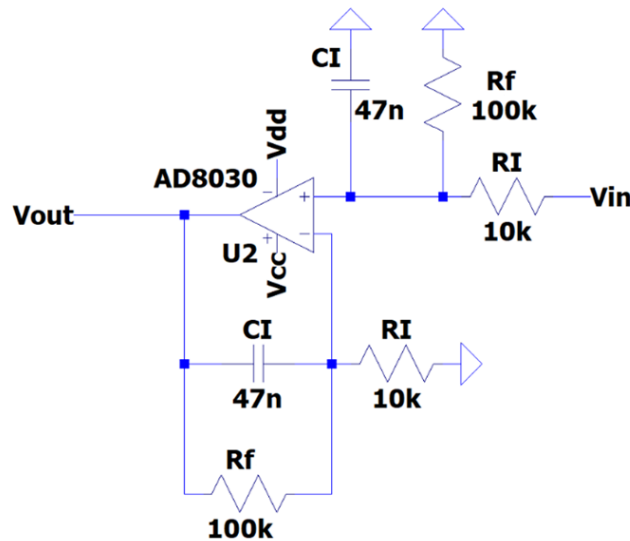


Figure 11: Final integral stage

Derivative Stage

The derivative action improves the dynamic behavior of the system by anticipating future variations of the error.

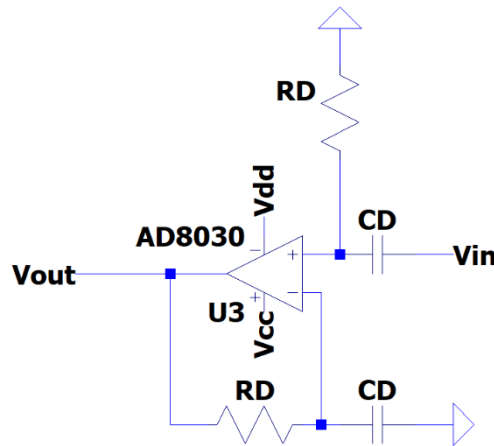


Figure 12: Differential stage

Recall that the voltages at the inverting and non-inverting inputs of an op-amp are equal and that the input currents are negligible.

Under these assumptions:

$$I_C = C_D \frac{d(V_{in} - V_+)}{dt}, \quad I_R = \frac{V_+}{R_D}$$

$$V_+ = R_D C_D \frac{d(V_{in} - V_+)}{dt}$$

Similarly:

$$\frac{V_{out} - V_-}{R_D} = C_D \frac{dV_-}{dt}$$

Since:

$$V_- = V_+ = V$$

$$\frac{V_{out} - V}{R_D} = C_D \frac{dV}{dt}$$

$$V_{out} = R_D C_D \frac{dV}{dt} + V$$

But:

$$V = R_D C_D \frac{d(V_{in} - V)}{dt}$$

Therefore:

$$R_D C_D \frac{dV_{in}}{dt} = V + R_D C_D \frac{dV}{dt}$$

The input RC network has a time constant:

$$\tau_D = R_D C_D$$

and a characteristic frequency:

$$f = \frac{1}{2\pi R_D C_D}$$

When the signal varies much more slowly than τ_D , the capacitor has sufficient time to charge and behaves almost like an open circuit.

When the signal varies rapidly, the capacitor behaves almost like a short circuit.

To better understand this behavior, consider a sinusoidal input:

$$V_{in}(t) = A \sin(\omega t)$$

With:

$$\omega = 2\pi f$$

If ω is very small, the capacitor reactance is high:

$$X_C = \frac{1}{\omega C}$$

And, therefore, very little current flows.

The voltage at node V remains close to zero, and the circuit output is consequently very small.

If instead:

$$\omega \gg \frac{1}{R_D C_D}$$

the capacitor reactance becomes very small and effectively provides a near-direct connection between V_{in} and V_+ .

Consequently:

$$V_+ = V_{in}$$

and from the feedback branch:

$$V_{out} = R_D C_D \frac{dV_{in}}{dt} + V_{in}$$

At high frequencies, the derivative term grows with frequency while the proportional term remains constant.

Therefore, at high frequency:

$$V_{out} = R_D C_D \frac{dV_{in}}{dt}$$

and the circuit behaves essentially as a differentiator.

In our case:

- $R_D = 12\text{ k}\Omega$
- $C_D = 47\text{ nF}$

thus:

$$V_{out} = R_D C_D \frac{dV_{in}}{dt} = 0.00056 \frac{dV_{in}}{dt}$$

This value is sufficient to improve loop damping without making the system excessively sensitive to the residual ripple generated by the switching converter.

Since the LM317 already attenuates a large portion of the ripple produced by the LM2596-ADJ, the derivative action can be kept relatively moderate.

Non-Inverting Summing Amplifier

The outputs of the proportional, derivative, and integral blocks are combined by means of a non-inverting summing amplifier.

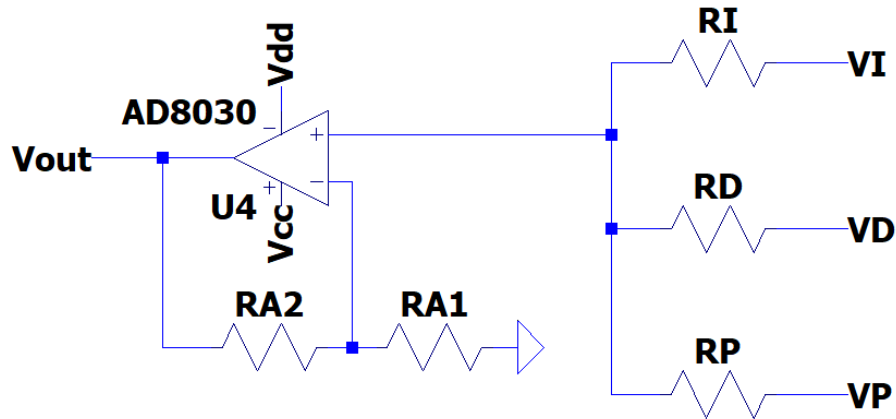


Figure 13: Summing stage

Each block can be weighted through its corresponding input resistor.

The op-amp is configured to provide an output voltage equal to:

$$V_{out} = \left(1 + \frac{R_{A2}}{R_{A1}}\right) \left[(V_P \frac{R_P}{R_P + \frac{R_D R_I}{R_D + R_I}}) + (V_I \frac{R_I}{R_I + \frac{R_D R_P}{R_D + R_P}}) + (V_D \frac{R_D}{R_D + \frac{R_P R_I}{R_P + R_I}}) \right]$$

Since:

- $R_P = 10k\Omega$
- $R_I = 33k\Omega$
- $R_D = 10k\Omega$

- $R_{A1} = 10k\Omega$
- $R_{A2} = 22k\Omega$

we obtain:

$$V_{out} = 3.2 \left[(V_P \frac{10}{10 + 7.6}) + (V_I \frac{33}{33 + 5}) + (V_D \frac{10}{10 + 7.6}) \right]$$

$$V_{out} = 1.8 V_P + 2.8 V_I + 1.8 V_D$$

Bandwidth Selection

Since the buck converter switches at approximately 150 kHz, there is no need for the PID controller to operate at high frequencies.

The purpose of the controller is not to regulate the internal switching cycle of the buck converter, which is already handled by the LM2596-ADJ, but rather to slowly adjust the average output voltage of the preregulator.

5. Complete Circuit and Simulations

After analyzing the LM2596-ADJ buck converter, the LM317 linear regulator, and the analog PID controller individually, it is now possible to examine the operation of the complete system. The resulting circuit combines a high-efficiency switching stage with a precision linear stage, creating a variable power supply capable of maintaining low power dissipation without sacrificing output voltage quality.

The proposed architecture is based on a dual-loop control structure. The first loop is the internal control loop of the LM2596-ADJ, responsible for regulating the output voltage of the buck converter. The second loop is implemented by the analog PID controller, which acts on the feedback node of the switching converter through a MOSFET. This outer loop does not directly control the final output voltage of the power supply; instead, it regulates the pre-regulation voltage supplied to the LM317. In this way, the buck converter becomes an adaptive voltage source that continuously adjusts its operating point according to the required output voltage.

The objective of the controller is to maintain an approximately constant voltage difference between the output of the switching converter and the final output of the power supply. Under ideal conditions, the system tends to satisfy the relationship:

$$V_{SW} = V_{out} + 5V$$

thereby providing the linear regulator with the voltage margin required for proper operation while simultaneously minimizing power dissipation.

The following figures show the complete power supply circuit and the results of a simulation in which the LM317 feedback resistance was varied to modify the output voltage. The purpose of the simulation is to verify that the control loop correctly adapts the output voltage of the switching pre-regulator, maintaining it approximately 5 V above the LM317 output voltage over the entire operating range.

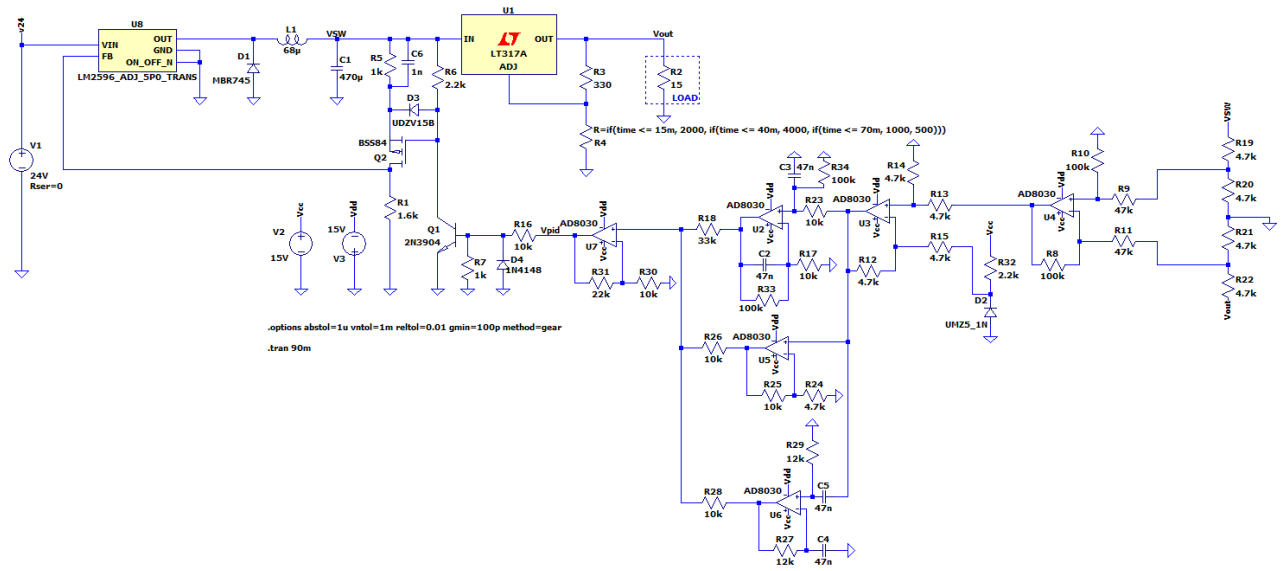


Figure 14: Complete circuit

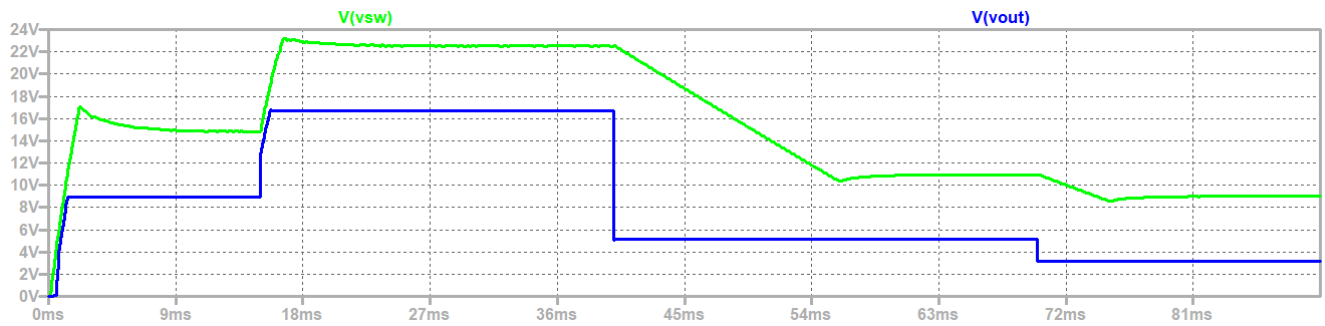


Figure 15: Voltages waveforms

6. Conclusion

This tutorial has presented the design of a variable power supply based on a hybrid architecture that combines an LM2596-ADJ buck converter with an LM317 linear regulator. The primary objective of the project was to combine the high efficiency typical of switching power supplies with the simplicity and regulation quality offered by linear regulators.

The theoretical analysis demonstrated how the use of a dynamic switching pre-regulator can significantly reduce the power dissipated by the linear regulator. In a conventional configuration, the LM317 must absorb the entire voltage difference between the input and output voltages, resulting in substantial power losses and significant thermal-management challenges. In the proposed circuit, however, the buck converter continuously adapts its output voltage according to the voltage demanded by the load, maintaining an approximately constant voltage margin across the linear regulator. This approach drastically reduces power dissipation, improves overall system efficiency, and simplifies thermal management.

Particular attention was devoted to the analog PID controller, which represents the key element of the entire project. Through the combined action of the proportional, integral, and derivative terms, the system can automatically regulate the pre-regulation voltage and adapt to changes in the operating point. The implementation using operational amplifiers and discrete components also provided an opportunity to explore fundamental aspects of control theory, highlighting practical issues such as coefficient tuning, control-loop stability, and the management of integrator windup.

The analysis of the interface stage between the PID controller and the switching converter demonstrated how the LM2596-ADJ output voltage can be modified without directly interfering with its internal control loop. The use of a PMOS transistor operating as a voltage-controlled resistor and driven by an NPN transistor makes it possible to continuously alter the converter's feedback divider, achieving simple and effective analog regulation of the preregulation voltage.

The simulations performed using LTspice proved to be an essential tool for validating the design. Simulation also enabled optimization of the PID controller

parameters before practical implementation, reducing development time and minimizing the number of modifications required on the prototype.

Although the circuit is more complex than a traditional linear power supply, the benefits obtained in terms of efficiency, reduced thermal dissipation, and regulation flexibility fully justify the adopted approach. The project also represents an interesting example of integration between classical analog techniques and modern switching converters, demonstrating that high performance can be achieved using widely available and relatively inexpensive components.

The architecture presented here can serve as a valuable starting point for further developments. Possible future improvements include the implementation of digital control algorithms, automatic PID tuning techniques, current-limiting functions, programmable output-voltage profiles, and advanced monitoring features based on microcontrollers or embedded systems. These enhancements could further increase the flexibility, efficiency, and functionality of the proposed power-supply architecture.